

Portfolio Management and Risk Consultation

FINSTAD Case 2 – Group 3 (Charlie)

CRUZ, Ricardo Miguel Iñigo GALEDO, Enrique Lorenzo Hermoso
SEBALLOS, Josiah Dweyn Panganiban SEECHUNG, Camille Castro
SISON, Aaron Joshua Estacio

2026-08-17

Table of contents

1	Setup and Library Loading	2
2	Executive Summary	2
2.0.1	Key Empirical Findings	3
2.0.2	Executive Recommendation	3
3	Introduction	3
4	Data Ingestion and Preparation	4
4.1	Daily Log Return Computation	5
4.2	Data Validation and Summary Check	6
5	Exploratory Financial Analysis	7
5.1	Quantitative Design System (QVRS Theme & Colors)	9
5.2	Historical Price & Return Trajectories	9
5.3	Distributional Shape: Histograms and Empirical Density vs Normal Reference	12
5.4	Quantile-Quantile (Q-Q) Tail Diagnostics	14
5.5	Dispersion of Daily Returns and Outlier Diagnostics	15
5.6	Cumulative Performance Comparison (Growth of \$1.00)	16
5.7	Correlation Structure	17
6	Portfolio Construction (Equal-Weight vs Monthly Rebalanced)	18
6.1	Comparative Performance Evaluation	20
6.2	Calendar-Year Return Decomposition	20
6.3	Cumulative Portfolio Growth Comparison	21
6.4	Portfolio Asset Weight Drift Analysis	23
7	Portfolio Optimization (Efficient Frontier, GMV, Max Sharpe)	24

8 Portfolio Risk Assessment (VaR, CVaR, Monte Carlo, Stress Testing)	30
9 Executive Investment Recommendation	37
10 References	38
11 Appendix: Complete R Analysis Code	39

1 Setup and Library Loading

Load the quantitative finance, portfolio analytics, and data visualization libraries required for the analysis.

```
# Core data manipulation and visualization
library(tidyverse)
library(lubridate)
library(scales)

# Time series and quantitative financial analysis
library(xts)
library(zoo)
library(PerformanceAnalytics)

# Solvers for portfolio optimization
library(nloptr)
library(quadprog)

# For Portfolio Risk Assessment
library(MASS)

# Table formatting and visual aesthetics
library(corrplot)
library(knitr)
library(kableExtra)
```

2 Executive Summary

This quantitative consultation report evaluates portfolio construction, optimization, and risk management strategies for Group 3 (Charlie) over the six-year sample period from January 2, 2020 through December 31, 2025. The mandated multi-asset investment universe comprises three Philippine equities (**TEL**, **MER**, **AEV**), two US international tech equities (**NVDA**, **META**), and one cryptocurrency (**BTC**), benchmarked against the S&P 500 ETF Trust (**SPY**) and evaluated against a 5.25% risk-free rate (Philippine 91-Day Treasury Bill).

2.0.1 Key Empirical Findings

1. **Portfolio Construction & Rebalancing Discipline:** An un-rebalanced Buy-and-Hold equal-weighted strategy delivered a cumulative return of 743.04% (42.83% CAGR) driven by massive asset weight drift into NVDA (61.75% of the portfolio by December 2025) and BTC (24.77%). Monthly rebalancing cut annualized volatility from 35.12% to 24.18% and contained the peak drawdown from -58.21% to -43.20%, at the cost of lower cumulative return (384.85%, 30.21% CAGR); the Sharpe ratio was marginally lower under rebalancing (1.0325 versus 1.0700).
2. **Mean-Variance Portfolio Optimization:**
 - **Global Minimum Variance (GMV):** Allocates 79.09% of capital to domestic Philippine equities (TEL: 31.01%, MER: 31.23%, AEV: 16.85%) with modest international tech exposure (NVDA: 3.15%, META: 12.86%) and 4.90% in BTC. GMV achieves an annualized return of **16.03%**, volatility of **20.22%**, and a Sharpe ratio of **0.5332**.
 - **Maximum Sharpe Ratio (Tangency Portfolio):** Concentrates in high-performing international growth engines (NVDA: 55.44%, BTC: 26.14%) plus MER (18.42%), with zero weights in TEL, AEV, and META, yielding an annualized return of **58.10%**, volatility of **38.25%**, and a Sharpe ratio of **1.3815**.
3. **Tail Risk & Stress Testing Resilience:**
 - Daily 95% Historical Value at Risk (VaR) is **-1.80%** (GMV) and **-3.39%** (Max Sharpe), while 95% Historical Expected Shortfall (CVaR) reaches **-2.85%** (GMV) and **-5.11%** (Max Sharpe).
 - Under historical crisis simulations (2020 COVID Crash, 2022 Crypto Winter, 2022 Fed Rate Hike), the GMV portfolio demonstrates resilience with a maximum historical shock loss of -22.55%, whereas the Max Sharpe portfolio incurs severe instantaneous drawdowns (-28.83% in the COVID crash, -45.30% in a 1.5x Black Swan event).

2.0.2 Executive Recommendation

We recommend the **Maximum Sharpe portfolio** (NVDA: 55.44%, BTC: 26.14%, MER: 18.42%) for investors with a high tolerance for volatility and drawdown, executed with **monthly systematic rebalancing**. With an expected return of **58.10%**, volatility of **38.25%**, and a Sharpe ratio of **1.3815** at a 5.25% risk-free rate, it is the most attractive risk-adjusted allocation in the universe. For investors prioritizing capital preservation, the **GMV portfolio** (TEL: 31.01%, MER: 31.23%, AEV: 16.85%, NVDA: 3.15%, META: 12.86%, BTC: 4.90%) is the defensive alternative, with 20.22% volatility, a daily 95% Historical VaR of -1.80%, and a maximum historical stress loss of -22.55%.

3 Introduction

The 2020–2025 sample period presented unprecedented structural shifts across global asset markets. Investors confronted the shock of the COVID-19 pandemic, massive fiscal and monetary stimulus, subsequent multi-decade inflation surges, aggressive monetary policy tightening

by the Federal Reserve and Bangko Sentral ng Pilipinas (BSP), the rapid commercialization of artificial intelligence, and the institutional integration of digital assets.

Navigating this regime requires rigorous quantitative portfolio management. Modern Portfolio Theory (MPT), pioneered by Markowitz (1952) and extended by Sharpe (1964), provides the foundational framework for optimizing the trade-off between expected return and portfolio variance. However, empirical financial returns exhibit pronounced non-Gaussian characteristics—specifically heavy tails (leptokurtosis) and volatility clustering—demanding advanced downside risk analytics including Value at Risk (VaR), Expected Shortfall (CVaR), Monte Carlo simulation, and historical stress testing (Jorion, 2007; Artzner et al., 1999).

This report delivers a comprehensive portfolio consultation for Group 3 (Charlie). Section 2 details data acquisition, cleaning, and log return calculations. Section 3 presents exploratory financial analysis and asset distribution diagnostics. Section 4 evaluates equal-weighted buy-and-hold versus monthly rebalanced execution. Section 5 solves for the Efficient Frontier, GMV, and Max Sharpe portfolios. Section 6 conducts quantitative risk assessments, and Section 7 formulates strategic investment recommendations.

4 Data Ingestion and Preparation

Load the compiled master dataset containing historical daily close prices for Group 3's asset universe (TEL, MER, AEV, NVDA, META, BTC, SPY) and the risk-free rate benchmark over the sample period (January 2, 2020 – December 31, 2025).

```
# Define path to master dataset
data_path <- file.path("../", "data", "master_dataset.csv")

if (!file.exists(data_path)) {
  data_path <- file.path("data", "master_dataset.csv")
}

raw_data <- read_csv(data_path, show_col_types = FALSE)

# Filter dataset strictly to sample period Jan 2, 2020 to Dec 31, 2025
raw_data <- raw_data %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2020-01-02") & Date <= as.Date("2025-12-31")) %>%
  arrange(Date)

# Preview structure
glimpse(raw_data)
```

Rows: 1,508

Columns: 9

```
$ Date      <date> 2020-01-02, 2020-01-03, 2020-01-06, 2020-01-07, 2020-01-08,
  ↪ 2~
```

```

$ TEL      <dbl> 12.94961, 13.00212, 12.92992, 13.35654, 13.46156, 13.75691,
  ↪ 13~
$ MER      <dbl> 6.232742, 6.173474, 6.090038, 6.039285, 5.919388, 5.967083,
  ↪ 6.~
$ AEV      <dbl> 1.0305720, 1.0345115, 1.0339358, 1.0260920, 1.0169658,
  ↪ 1.02689~
$ NVDA     <dbl> 5.963804, 5.868347, 5.892957, 5.964300, 5.975487, 6.041113,
  ↪ 6.~
$ META     <dbl> 207.9538, 206.8535, 210.7493, 211.2053, 213.3465, 216.3997,
  ↪ 21~
$ BTC      <dbl> 6985.470, 7344.884, 7769.219, 8163.692, 8079.863, 7879.071,
  ↪ 81~
$ SPY      <dbl> 296.1253, 293.8830, 295.0042, 294.1747, 295.7425, 297.7478,
  ↪ 29~
$ RF_Rate  <dbl> 0.0525, 0.0525, 0.0525, 0.0525, 0.0525, 0.0525, 0.0525,
  ↪ 0.0525~

```

4.1 Daily Log Return Computation

Compute daily log returns for all portfolio assets and the benchmark:

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

```

prices_df <- raw_data

# Extract asset price columns
asset_cols <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC", "SPY")

# Convert matrix to xts time series object
prices_mat <- as.matrix(prices_df[, asset_cols])
prices_xts <- xts(prices_mat, order.by = prices_df[["Date"]])

returns_xts <- Return.calculate(prices_xts, method = "log")[-1, ]

# Separate portfolio assets from benchmark
portfolio_assets <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC")
benchmark_asset <- "SPY"

asset_returns_xts <- returns_xts[, portfolio_assets]
benchmark_returns_xts <- returns_xts[, benchmark_asset]

# No-trade days carry the last price mark, so the resumption-day return
# captures the full move across the gap. Wealth indexes use exp(cumsum(r))
# because these are log returns; cumprod(1 + r) would misstate growth.

```

```

prices_carried <- na.locf(prices_xts)
returns_xts_carried <- Return.calculate(prices_carried, method = "log")[-1,
  ↪ ]
asset_returns_carried <- returns_xts_carried[, portfolio_assets]

# EDA series: carried returns with no-trade days set back to NA, so
# resumption-day returns (full gap moves) enter the moments, correlations,
# and distribution figures while no-trade days stay missing.
returns_xts_eda <- returns_xts_carried
returns_xts_eda[is.na(prices_xts[-1, ])] <- NA
asset_returns_eda <- returns_xts_eda[, portfolio_assets]

# Risk-free rate (convert annual T-Bill rate to daily log-equivalent rate)
rf_annual <- mean(prices_df[["RF_Rate"]], na.rm = TRUE)
rf_daily <- log(1 + rf_annual) / 252

cat("Dataset span:", format(min(index(returns_xts))), "to",
  ↪ format(max(index(returns_xts))), "\n")

```

Dataset span: 2020-01-03 to 2025-12-31

```
cat("Total trading days:", nrow(returns_xts), "\n")
```

Total trading days: 1507

```
cat("Annualized Risk-Free Rate:", percent(rf_annual, accuracy = 0.01), "\n")
```

Annualized Risk-Free Rate: 5.25%

```
cat("Daily Log Risk-Free Rate:", percent(rf_daily, accuracy = 0.0001), "\n")
```

Daily Log Risk-Free Rate: 0.0203%

4.2 Data Validation and Summary Check

Verify data integrity, check for missing values, and print head records of returns.

```

missing_summary <- colSums(is.na(returns_xts))
notrade_summary <- colSums(is.na(returns_xts_eda))
cat("Missing return days per asset (raw gap-boundary convention):\n")

```

Missing return days per asset (raw gap-boundary convention):

```
print(missing_summary)
```

TEL	MER	AEV	NVDA	META	BTC	SPY
0	162	163	0	0	0	0

Table 1: First 6 Days of Asset Daily Log Returns (2020)

Date	TEL	MER	AEV	NVDA	META	BTC	SPY
2020-01-03	0.0040	-0.0096	0.0038	-0.0161	-0.0053	0.0502	-0.0076
2020-01-06	-0.0056	-0.0136	-0.0006	0.0042	0.0187	0.0562	0.0038
2020-01-07	0.0325	-0.0084	-0.0076	0.0120	0.0022	0.0495	-0.0028
2020-01-08	0.0078	-0.0201	-0.0089	0.0019	0.0101	-0.0103	0.0053
2020-01-09	0.0217	0.0080	0.0097	0.0109	0.0142	-0.0252	0.0068
2020-01-10	0.0029	0.0189	-0.0002	0.0053	-0.0011	0.0358	-0.0029

```
cat("No-trade days per asset (gap-aware convention):\n")
```

No-trade days per asset (gap-aware convention):

```
print(notrade_summary)
```

```
TEL  MER  AEV  NVDA  META  BTC  SPY
  0   86   87   0    0    0    0
```

```
if (any(missing_summary > 0)) {
  message("Missing return days present; verify no-trade day treatment in
    ↪ data acquisition")
}
```

```
# Format first 6 rows for table display
returns_head_df <- as.data.frame(head(returns_xts)) %>%
  rownames_to_column(var = "Date") %>%
  mutate(across(-Date, ~ sprintf("%.4f", .x)))
```

```
kbl(returns_head_df, booktabs = TRUE, align = "c") %>%
  kable_styling(latex_options = c("striped", "hold_position"),
    ↪ bootstrap_options = c("striped", "hover"))
```

5 Exploratory Financial Analysis

```
stats_mat <- table.Stats(asset_returns_eda)

# Convert to data frame for clean rendering
stats_df <- as.data.frame(stats_mat) %>%
  rownames_to_column(var = "Metric")

kbl(stats_df, booktabs = TRUE, digits = 4) %>%
```

Table 2: Descriptive Statistics for Portfolio Assets (Jan 2, 2020 – Dec 31, 2025)

Metric	TEL	MER	AEV	NVDA	META	BTC
Observations	1507.0000	1421.0000	1420.0000	1507.0000	1507.0000	1507.0000
NAs	0.0000	86.0000	87.0000	0.0000	0.0000	0.0000
Minimum	-0.2703	-0.1668	-0.1680	-0.2040	-0.3064	-0.4647
Quartile 1	-0.0098	-0.0106	-0.0141	-0.0157	-0.0117	-0.0161
Median	0.0000	-0.0001	0.0003	0.0032	0.0010	0.0010
Arithmetic Mean	0.0003	0.0003	-0.0005	0.0023	0.0008	0.0017
Geometric Mean	0.0001	0.0001	-0.0009	0.0017	0.0004	0.0009
Quartile 3	0.0101	0.0112	0.0127	0.0215	0.0142	0.0202
Maximum	0.1548	0.1175	0.1258	0.2181	0.2093	0.1915
SE Mean	0.0005	0.0005	0.0007	0.0009	0.0007	0.0010
LCL Mean (0.95)	-0.0007	-0.0007	-0.0019	0.0006	-0.0006	-0.0003
UCL Mean (0.95)	0.0013	0.0014	0.0008	0.0040	0.0022	0.0037
Variance	0.0004	0.0004	0.0006	0.0011	0.0008	0.0015
Stdev	0.0199	0.0204	0.0253	0.0333	0.0279	0.0393
Skewness	-1.7677	-0.2346	-0.2215	0.0439	-1.0740	-1.2216
Kurtosis	31.4634	5.7962	3.7706	4.0527	22.0895	16.2926

```
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
  → bootstrap_options = c("striped", "hover"))
```

```
benchmark_stats_df <- as.data.frame(table.Stats(returns_xts_eda[, "SPY"]))
→ %>%
  rownames_to_column(var = "Metric") %>%
  filter(Metric %in% c("Observations", "Arithmetic Mean", "Stdev",
    → "Skewness", "Kurtosis", "Minimum", "Maximum")) %>%
  mutate(Metric = recode(Metric, "Arithmetic Mean" = "Mean"))

safe_kable(benchmark_stats_df, caption = "SPY Benchmark Descriptive
→ Statistics (Daily Log Returns, Jan 2, 2020 -- Dec 31, 2025)")
```

Table 3: SPY Benchmark Descriptive Statistics (Daily Log Returns, Jan 2, 2020 – Dec 31, 2025)

Metric	SPY
Observations	1507.0000
Minimum	-0.1159
Mean	0.0005
Maximum	0.0999
Stdev	0.0131
Skewness	-0.5584

5.1 Quantitative Design System (QVRS Theme & Colors)

```
# QVRS Economist Color Tokens
qvrs_colors <- c(
  "TEL" = "#1F2E7A", # Chicago 30
  "MER" = "#2E45B8", # Chicago 45
  "AEV" = "#475ED1", # Chicago 55
  "NVDA" = "#1DC9A4", # Hong Kong 45
  "META" = "#F97A1F", # Singapore 55
  "BTC" = "#E3120B", # Economist Red
  "SPY" = "#595959" # London 35
)

# Custom Economist/QVRS ggplot2 theme
theme_quant <- function() {
  theme_minimal() +
    theme(
      text = element_text(color = "#0D0D0D"),
      plot.title = element_text(face = "bold", size = 12, margin = margin(b
        → = 4)),
      plot.subtitle = element_text(size = 9.5, color = "#333333", margin =
        → margin(b = 8)),
      plot.caption = element_text(size = 8, color = "#595959", hjust = 0,
        → margin = margin(t = 6)),
      axis.title = element_text(size = 9, face = "bold", color = "#1A1A1A"),
      axis.text = element_text(size = 8, color = "#333333"),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "#F2F2F2", linewidth = 0.3),
      legend.position = "bottom",
      legend.title = element_blank(),
      axis.ticks = element_blank(),
      strip.background = element_rect(fill = "#EBEDFA", color = NA),
      strip.text = element_text(face = "bold", size = 9, color = "#141F52")
    )
}
```

5.2 Historical Price & Return Trajectories

```
# Compute normalized price index (Base = 100 at start of sample period)
prices_norm_df <- as.data.frame(prices_carried) %>%
  rownames_to_column(var = "Date") %>%
```

```

mutate(Date = as.Date(Date)) %>%
pivot_longer(-Date, names_to = "Asset", values_to = "Price") %>%
group_by(Asset) %>%
mutate(Normalized_Price = (Price / first(Price)) * 100) %>%
ungroup() %>%
mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(prices_norm_df, aes(x = Date, y = Normalized_Price, color = Asset)) +
  geom_line(linewidth = 0.75, alpha = 0.9) +
  scale_color_manual(values = qvrs_colors) +
  scale_y_log10(labels = label_comma(prefix = "$")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Historical Normalized Asset Price Trajectories (Log Scale)",
    subtitle = "Base = 100 on January 2, 2020 through December 31, 2025",
    x = NULL,
    y = "Normalized Index (Jan 2020 = 100)"
  ) +
  theme_quant()

```

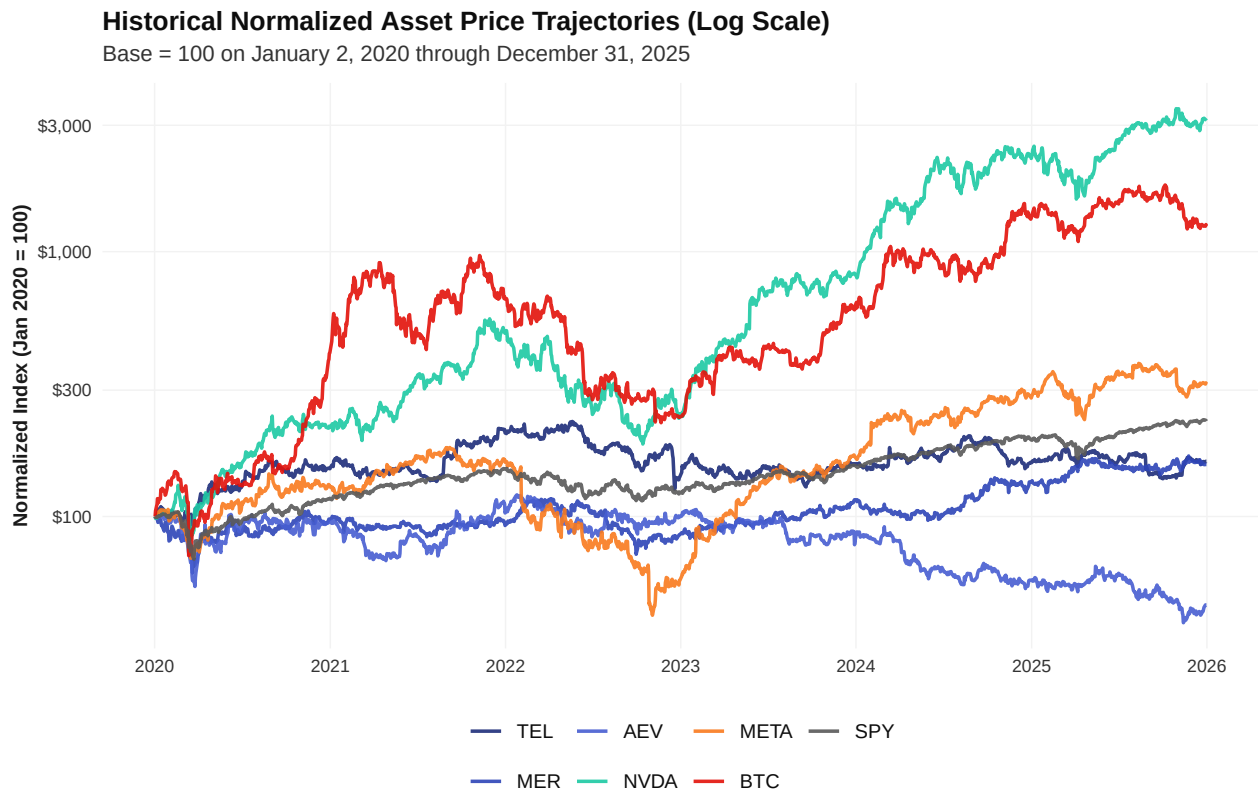


Figure 1: Historical Normalized Asset Price Trajectories (Base = 100 on Jan 2, 2020)

```

# Reshape returns for tidy plotting
returns_tidy <- as.data.frame(returns_xts_eda) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Return") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(returns_tidy, aes(x = Date, y = Return, color = Asset)) +
  geom_line(linewidth = 0.35, alpha = 0.85) +
  facet_wrap(~ Asset, ncol = 2, scales = "free_y") +
  scale_color_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_x_date(date_labels = "%Y", date_breaks = "2 years") +
  labs(
    title = "Daily Log Return Dynamics and Volatility Clustering",
    subtitle = "Sample Period: January 2, 2020 through December 31, 2025",
    x = NULL,
    y = "Daily Log Return"
  ) +
  theme_quant() +
  theme(legend.position = "none")

```

Daily Log Return Dynamics and Volatility Clustering

Sample Period: January 2, 2020 through December 31, 2025

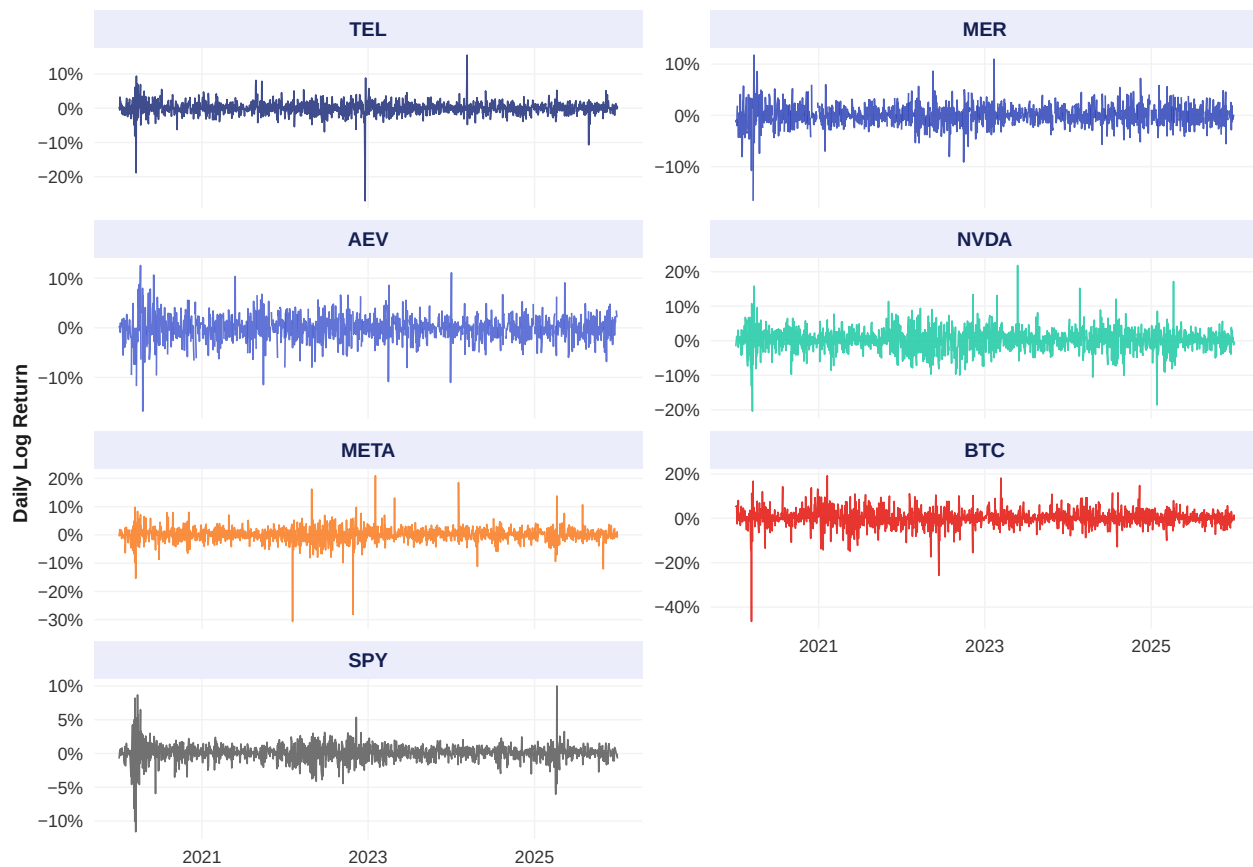


Figure 2: Daily Log Return Trajectories Across Portfolio Universe (Jan 2020 – Dec 2025)

5.3 Distributional Shape: Histograms and Empirical Density vs Normal Reference

```
# Compute normal distribution parameters per asset for reference curves
norm_params <- returns_tidy %>%
  group_by(Asset) %>%
  summarise(
    mean = mean(Return, na.rm = TRUE),
    sd = sd(Return, na.rm = TRUE),
    .groups = "drop"
  )

# Grid of x values for theoretical normal density overlay
grid_df <- norm_params %>%
  group_by(Asset) %>%
  reframe(
    x = seq(mean - 4 * sd, mean + 4 * sd, length.out = 200),
```

```

    y = dnorm(x, mean = mean, sd = sd)
  )

ggplot(returns_tidy, aes(x = Return)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 60,
    fill = "#D6DBF5",
    color = "#1F2E7A",
    linewidth = 0.2,
    alpha = 0.7
  ) +
  geom_density(aes(color = "Empirical Density"), linewidth = 0.8) +
  geom_line(
    data = grid_df,
    aes(x = x, y = y, color = "Gaussian Normal"),
    linetype = "dashed",
    linewidth = 0.75
  ) +
  facet_wrap(~ Asset, scales = "free", ncol = 3) +
  scale_color_manual(
    values = c("Empirical Density" = "#141F52", "Gaussian Normal" =
      ↪ "#E3120B")
  ) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Empirical Distribution Shape: Heavy Tails and Non-Normality",
    subtitle = "Comparison of empirical kernel density against theoretical
      ↪ normal distribution",
    x = "Daily Log Return",
    y = "Density"
  ) +
  theme_quant()

```

Empirical Distribution Shape: Heavy Tails and Non-Normality

Comparison of empirical kernel density against theoretical normal distribution

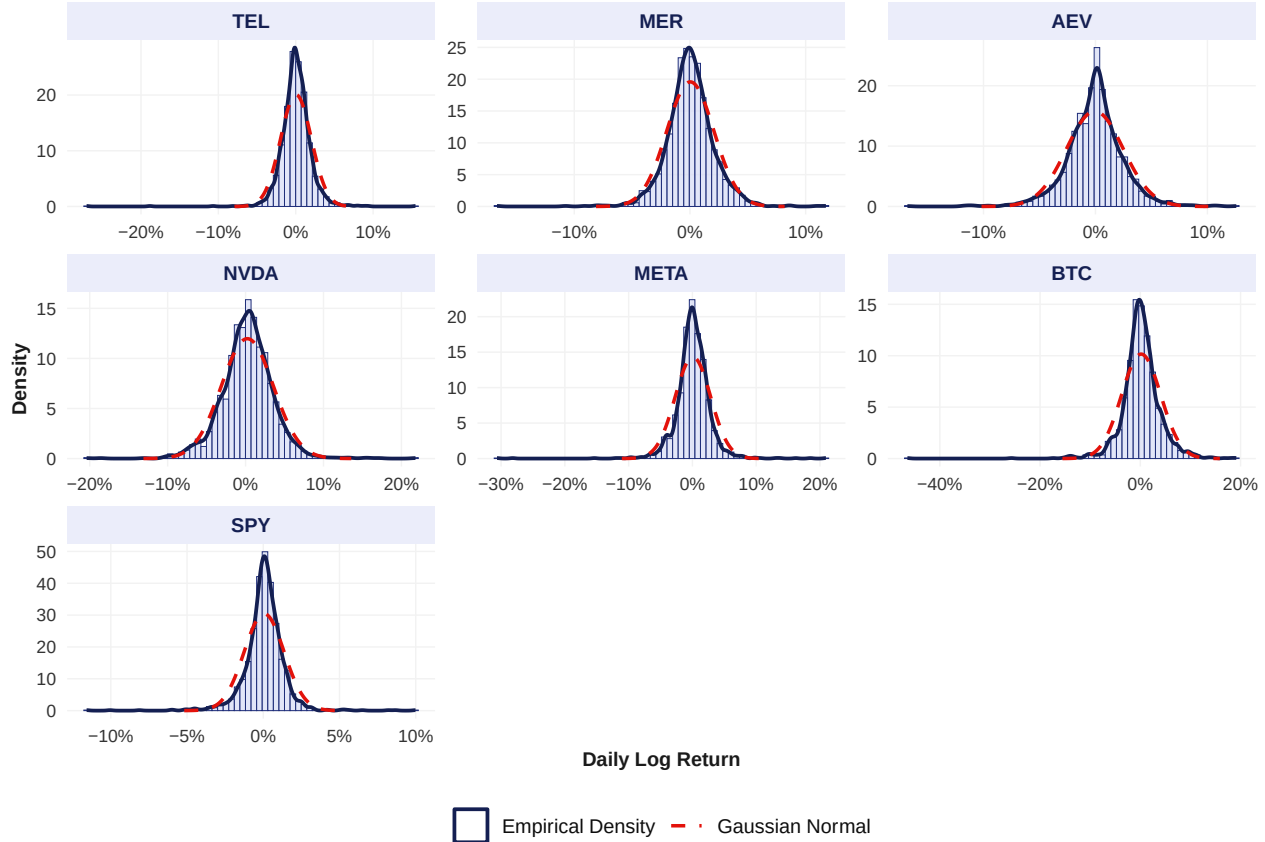


Figure 3: Empirical Return Distributions vs Gaussian Reference Curves

5.4 Quantile-Quantile (Q-Q) Tail Diagnostics

```
ggplot(returns_tidy, aes(sample = Return)) +
  stat_qq_line(color = "#E3120B", linewidth = 0.7, linetype = "dashed") +
  stat_qq(color = "#1F2E7A", alpha = 0.5, size = 0.8) +
  facet_wrap(~ Asset, scales = "free", ncol = 3) +
  labs(
    title = "Quantile-Quantile Tail Departure from Normality",
    subtitle = "S-shaped curves confirm pronounced fat-tail behavior across
      ↪ asset returns",
    x = "Theoretical Gaussian Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_quant()
```

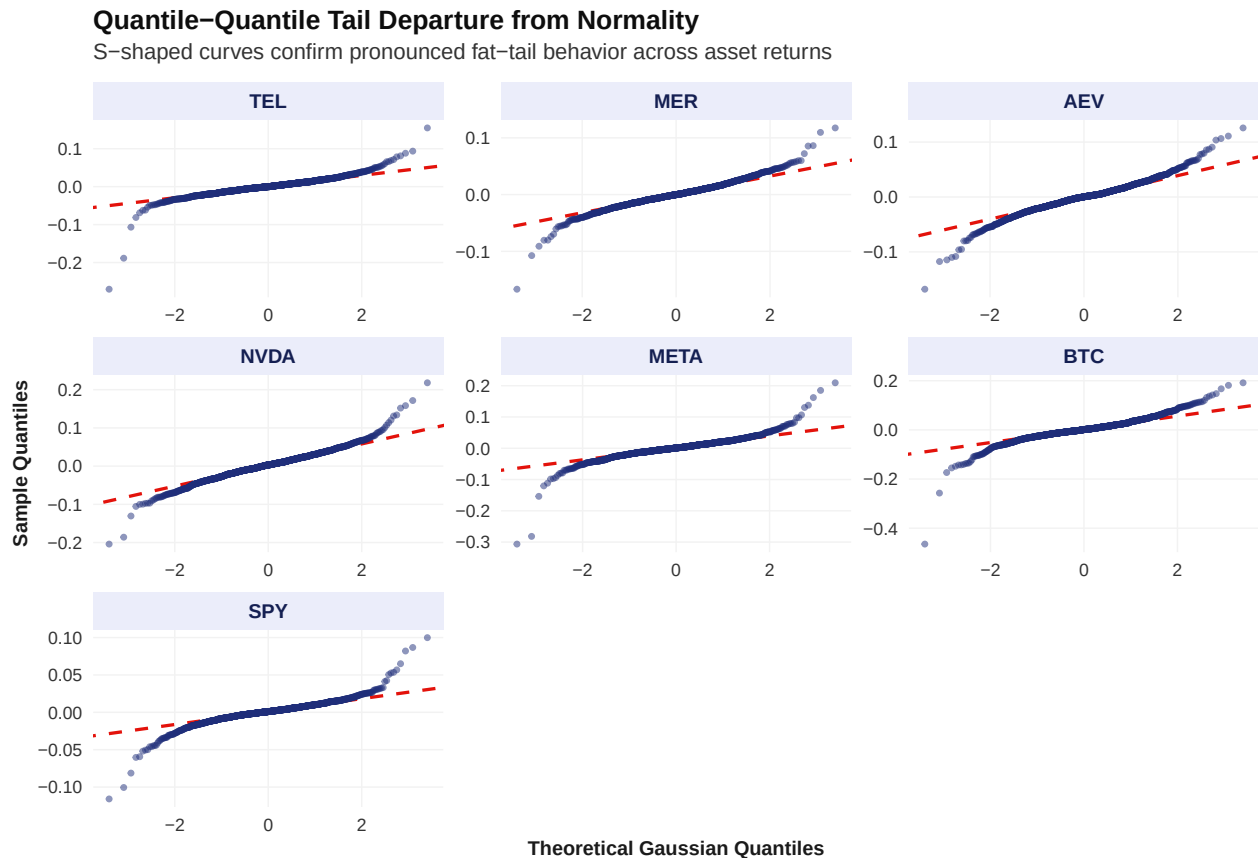


Figure 4: Normal Q-Q Diagnostics Demonstrating Heavy Tail Leptokurtosis

5.5 Dispersion of Daily Returns and Outlier Diagnostics

```
ggplot(returns_tidy, aes(x = Asset, y = Return, fill = Asset)) +
  geom_boxplot(outlier.color = "#E3120B", outlier.size = 1, outlier.alpha =
    ↪ 0.5, linewidth = 0.4) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Cross-Asset Return Dispersion and Extreme Tail Outliers",
    subtitle = "Comparison of median, interquartile range (IQR), and extreme
      ↪ shocks",
    x = "Asset",
    y = "Daily Log Return"
  ) +
  theme_quant() +
  theme(legend.position = "none")
```

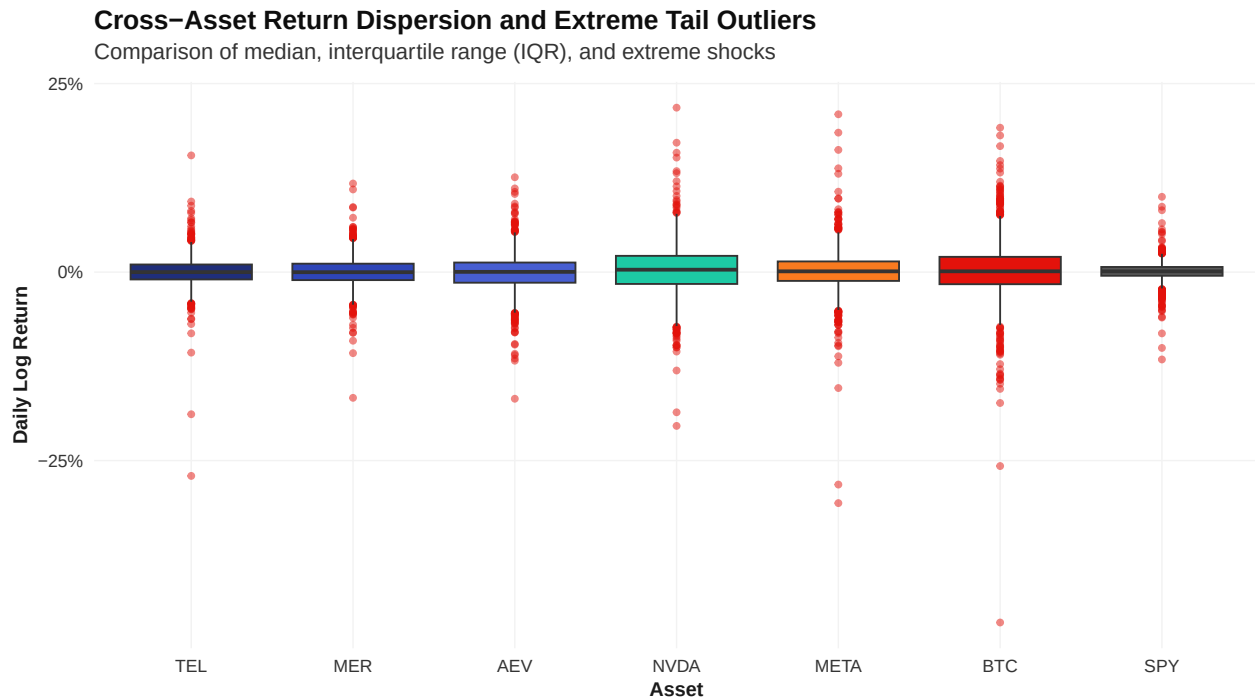


Figure 5: Return Dispersion and Extreme Outlier Boxplots

5.6 Cumulative Performance Comparison (Growth of \$1.00)

```
cum_returns_df <- as.data.frame(exp(cumsum(returns_xts_carried))) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Wealth") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(cum_returns_df, aes(x = Date, y = Wealth, color = Asset)) +
  geom_line(linewidth = 0.75) +
  scale_color_manual(values = qvrs_colors) +
  scale_y_log10(labels = label_dollar(prefix = "USD ")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Cumulative Wealth Index Growth Trajectories (Log Scale)",
    subtitle = "Hypothetical USD 1.00 investment performance from January
  ↪ 2020 to December 2025",
    x = NULL,
    y = "Wealth Index (USD)"
  ) +
  theme_quant()
```

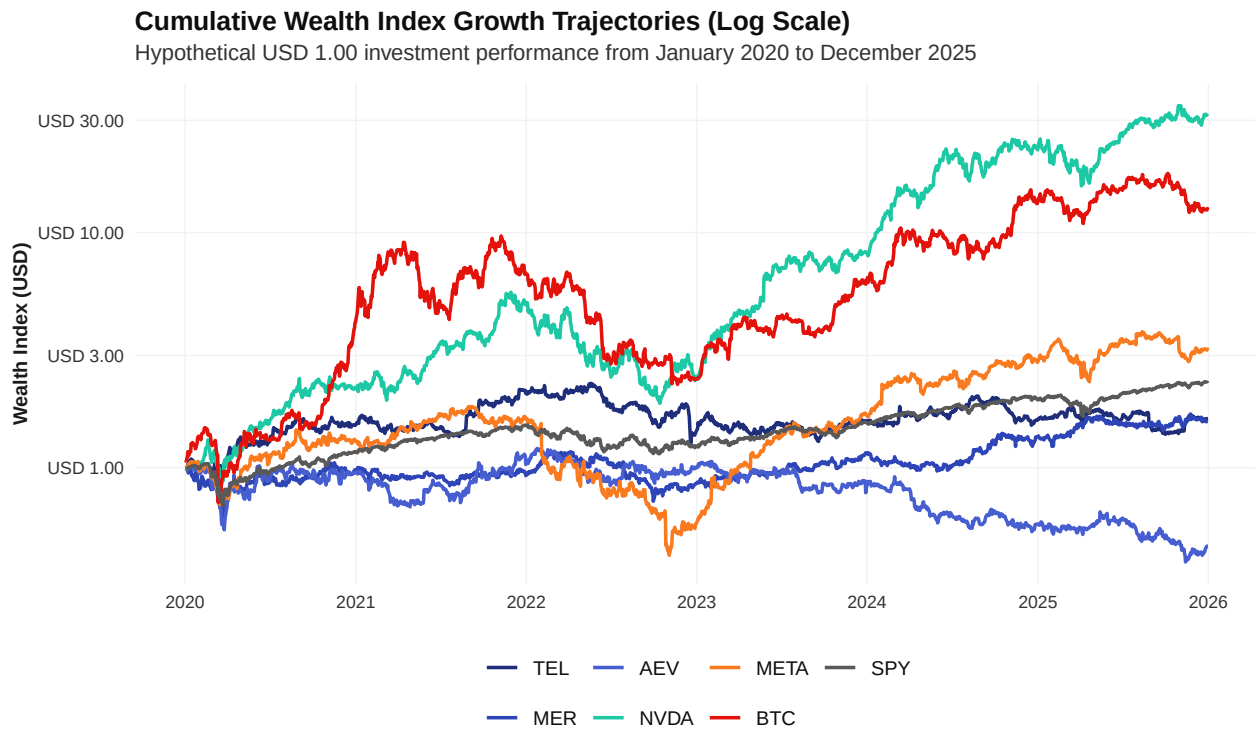



Figure 6: Cumulative Growth Trajectories of USD 1.00 Investment (Jan 2020 – Dec 2025)

5.7 Correlation Structure

```
cor_matrix <- cor(returns_xts_eda, use = "pairwise.complete.obs")

# Render correlation heatmap
corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  number.cex = 0.8,
  mar = c(0, 0, 1, 0)
)
```

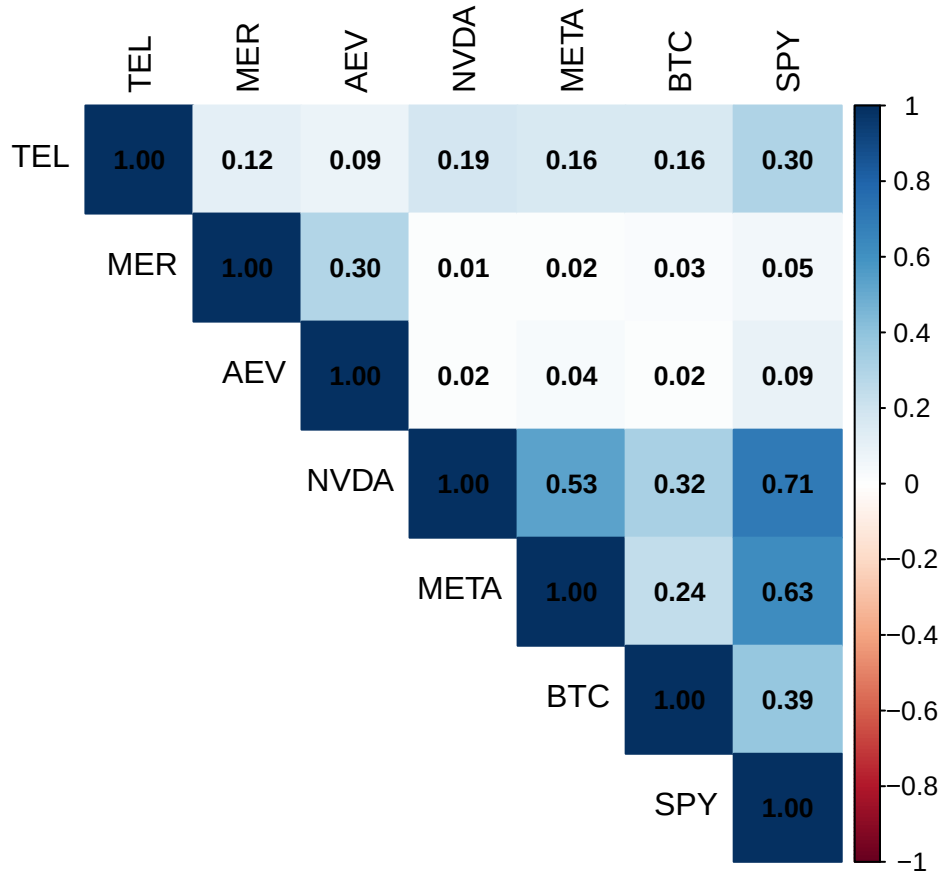


Figure 7: Asset Return Correlation Heatmap (Group 3 Universe)

6 Portfolio Construction (Equal-Weight vs Monthly Rebalanced)

This section evaluates the performance of equal-weighted asset allocations under two distinct execution regimes: a static Buy-and-Hold strategy and a disciplined Monthly Rebalanced strategy. The 6-asset portfolio universe (TEL, MER, AEV, NVDA, META, BTC) starts with equal baseline weights ($w_i = 1/6 \approx 16.67\%$) on January 2, 2020 and is benchmarked against the SPDR S&P 500 ETF Trust (SPY).

```
# Simple (arithmetic) returns: a weighted sum of log returns is not the
# portfolio return, so Return.portfolio and all portfolio metrics below
# operate on simple returns converted from the carried log series.
ret_smp <- exp(returns_xts_carried) - 1
asset_returns_smp <- ret_smp[, portfolio_assets]

# Equal weight allocation vector across the 6 portfolio assets
w_eq <- rep(1 / 6, 6)
names(w_eq) <- colnames(asset_returns_smp)
```

```

# Strategy 1: Buy-and-Hold Equal-Weighted Portfolio
port_bnh <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_eq,
  verbose = TRUE
)
r_bnh <- port_bnh$returns
weights_bnh <- port_bnh$EOP.Weight

# Strategy 2: Monthly Rebalanced Equal-Weighted Portfolio
port_reb <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_eq,
  rebalance_on = "months",
  verbose = TRUE
)
r_reb <- port_reb$returns
weights_reb <- port_reb$EOP.Weight

# Risk-free rate metrics (5.25% per annum)
rf_annual <- 0.0525
rf_daily <- log(1 + rf_annual) / 252

# Performance evaluation helper function
eval_portfolio <- function(r_series, name) {
  cum_ret <- as.numeric(tail(cumprod(1 + r_series), 1)) - 1
  n_obs <- length(r_series)
  ann_ret <- (1 + cum_ret)^(252 / n_obs) - 1
  ann_sd <- sd(as.numeric(r_series)) * sqrt(252)
  sharpe <- (ann_ret - rf_annual) / ann_sd
  max_dd <- as.numeric(maxDrawdown(r_series))

  data.frame(
    Strategy = name,
    `Cumulative Return` = sprintf("%.2f%%", cum_ret * 100),
    `CAGR (Annualized)` = sprintf("%.2f%%", ann_ret * 100),
    `Annual Volatility` = sprintf("%.2f%%", ann_sd * 100),
    `Sharpe Ratio (Rf=5.25%)` = sprintf("%.4f", sharpe),
    `Max Drawdown` = sprintf("%.2f%%", max_dd * 100),
    check.names = FALSE
  )
}

perf_summary_df <- rbind(
  eval_portfolio(r_bnh, "Equal-Weight (Buy & Hold)"),

```

```
eval_portfolio(r_reb, "Equal-Weight (Monthly Rebalanced)",
eval_portfolio(ret_smp[, "SPY"], "SPDR S&P 500 ETF (SPY Benchmark)")
)
```

6.1 Comparative Performance Evaluation

```
safe_kable(perf_summary_df, caption = "Portfolio Performance Summary
↪ Comparison (Jan 2, 2020 -- Dec 31, 2025)")
```

Table 4: Portfolio Performance Summary Comparison (Jan 2, 2020 – Dec 31, 2025)

Strategy	Cumulative Return	CAGR (Annualized)	Annual Volatility	Sharpe Ratio (Rf=5.25%)	Max
Equal-Weight (Buy & Hold)	743.04%	42.83%	35.12%	1.0700	58.21
Equal-Weight (Monthly Rebalanced)	384.85%	30.21%	24.18%	1.0325	43.20
SPDR S&P 500 ETF (SPY Benchmark)	129.06%	14.87%	20.75%	0.4635	33.72

6.2 Calendar-Year Return Decomposition

```
yearly_ret <- function(r_series, yr) {
  rr <- r_series[format(index(r_series), "%Y") == yr]
  as.numeric(tail(cumprod(1 + rr), 1)) - 1
}

cyr_strategy_df <- do.call(rbind, lapply(2020:2025, function(yr) {
  data.frame(
    Year = yr,
    `Buy & Hold` = sprintf("%.2f%%", yearly_ret(r_bnh, yr) * 100),
    `Monthly Rebalanced` = sprintf("%.2f%%", yearly_ret(r_reb, yr) * 100),
    `SPY Benchmark` = sprintf("%.2f%%", yearly_ret(ret_smp[, "SPY"], yr) *
↪ 100),
    check.names = FALSE
  )
}))

safe_kable(cyr_strategy_df, caption = "Calendar-Year Strategy Returns
↪ (Simple Returns, 2020--2025)")
```

Table 5: Calendar-Year Strategy Returns (Simple Returns, 2020–2025)

Year	Buy & Hold	Monthly Rebalanced	SPY Benchmark
2020	84.72%	70.37%	17.24%
2021	54.93%	46.80%	28.73%
2022	-49.57%	-36.77%	-18.18%

2023	125.47%	85.09%	26.18%
2024	116.05%	50.88%	24.89%
2025	19.92%	9.78%	17.72%

```
cyr_asset_df <- do.call(rbind, lapply(2020:2025, function(yr) {
  data.frame(
    Year = yr,
    TEL = sprintf("%.2f%%", yearly_ret(ret_smp[, "TEL"], yr) * 100),
    MER = sprintf("%.2f%%", yearly_ret(ret_smp[, "MER"], yr) * 100),
    AEV = sprintf("%.2f%%", yearly_ret(ret_smp[, "AEV"], yr) * 100),
    NVDA = sprintf("%.2f%%", yearly_ret(ret_smp[, "NVDA"], yr) * 100),
    META = sprintf("%.2f%%", yearly_ret(ret_smp[, "META"], yr) * 100),
    BTC = sprintf("%.2f%%", yearly_ret(ret_smp[, "BTC"], yr) * 100),
    check.names = FALSE
  )
}))

safe_kable(cyr_asset_df, caption = "Calendar-Year Asset Returns (Simple
↪ Returns, 2020--2025)")
```

Table 6: Calendar-Year Asset Returns (Simple Returns, 2020–2025)

Year	TEL	MER	AEV	NVDA	META	BTC
2020	51.43%	-2.20%	-4.29%	118.02%	30.21%	315.17%
2021	36.56%	-5.47%	7.75%	125.48%	23.13%	59.67%
2022	-31.70%	-7.63%	-3.29%	-50.26%	-64.22%	-64.15%
2023	11.92%	35.53%	-21.55%	239.02%	194.13%	153.57%
2024	0.81%	16.71%	-26.51%	171.25%	66.05%	121.93%
2025	1.48%	16.38%	-19.34%	38.92%	13.09%	-6.34%

6.3 Cumulative Portfolio Growth Comparison

```
# Merge portfolio return series (all simple returns)
port_returns_combined <- cbind(r_bnh, r_reb, ret_smp[, "SPY"])
colnames(port_returns_combined) <- c("Buy_Hold", "Monthly_Rebalanced",
↪ "SPY_Benchmark")

port_growth_df <- as.data.frame(cumprod(1 + port_returns_combined)) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Strategy", values_to = "Wealth") %>%
  mutate(
    Strategy = case_when(
      Strategy == "Buy_Hold" ~ "Equal-Weight (Buy & Hold)",
```

```

    Strategy == "Monthly_Rebalanced" ~ "Equal-Weight (Monthly
→ Rebalanced)",
    Strategy == "SPY_Benchmark" ~ "SPDR S&P 500 ETF (SPY)"
  ),
  Strategy = factor(
    Strategy,
    levels = c("Equal-Weight (Buy & Hold)", "Equal-Weight (Monthly
→ Rebalanced)", "SPDR S&P 500 ETF (SPY)")
  )
)

strat_colors <- c(
  "Equal-Weight (Buy & Hold)" = "#E3120B",          # Economist Red
  "Equal-Weight (Monthly Rebalanced)" = "#1F2E7A", # Chicago 30
  "SPDR S&P 500 ETF (SPY)" = "#595959"             # London 35
)

ggplot(port_growth_df, aes(x = Date, y = Wealth, color = Strategy, linetype
→ = Strategy)) +
  geom_line(linewidth = 0.85) +
  scale_color_manual(values = strat_colors) +
  scale_linetype_manual(values = c("solid", "solid", "dashed")) +
  scale_y_log10(labels = label_dollar(prefix = "USD ")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Cumulative Wealth Growth Comparison (Log Scale)",
    subtitle = "Performance of USD 1.00 allocation from January 2020 through
→ December 2025",
    x = NULL,
    y = "Wealth Index (USD)"
  ) +
  theme_quant()

```

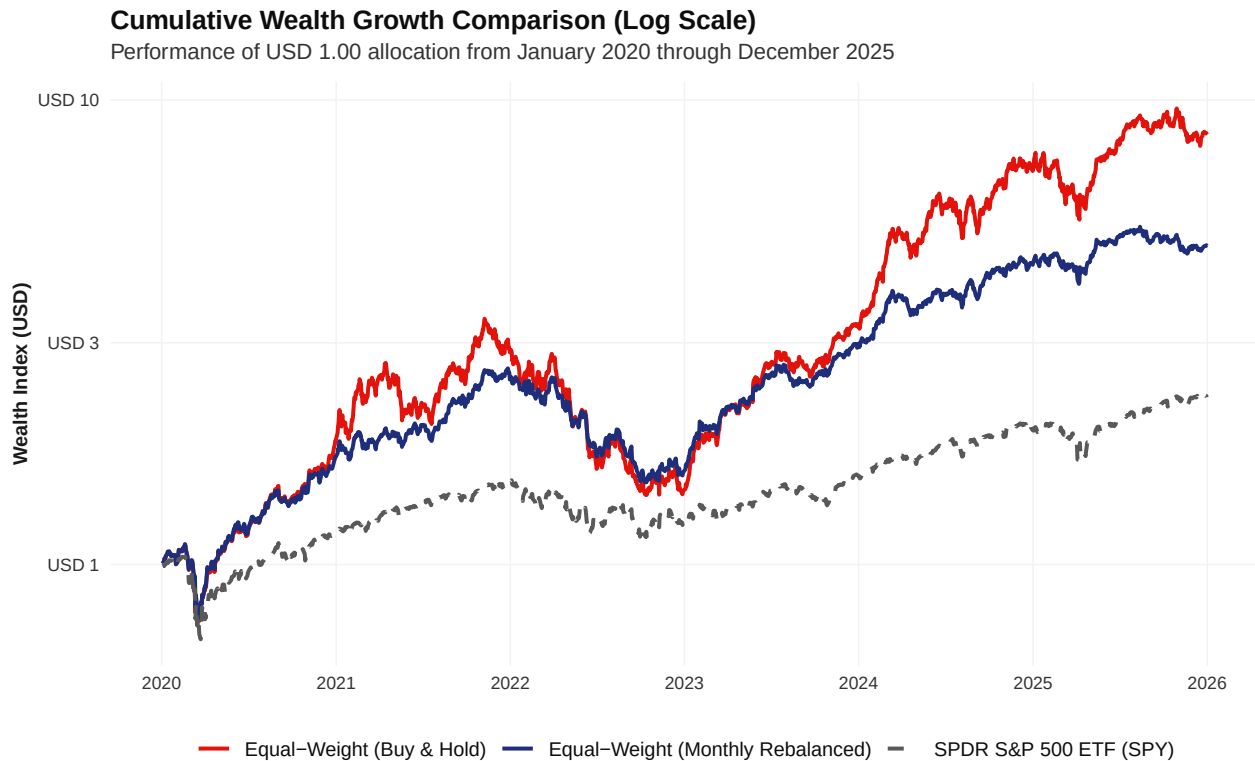


Figure 8: Cumulative Growth Trajectories of USD 1.00 Investment Across Strategies

6.4 Portfolio Asset Weight Drift Analysis

```
# Reshape end-of-period weights for Buy & Hold strategy
weights_bnh_df <- as.data.frame(weights_bnh) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Weight") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(weights_bnh_df, aes(x = Date, y = Weight, fill = Asset)) +
  geom_area(alpha = 0.85, color = "#FFFFFF", linewidth = 0.1) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Asset Allocation Drift in Buy-and-Hold Execution",
    subtitle = "Evolution of asset weights driven by differential market
    ↪ performance (Jan 2020 -- Dec 2025)",
    x = NULL,
    y = "Portfolio Weight Share"
  ) +
  theme_quant()
```

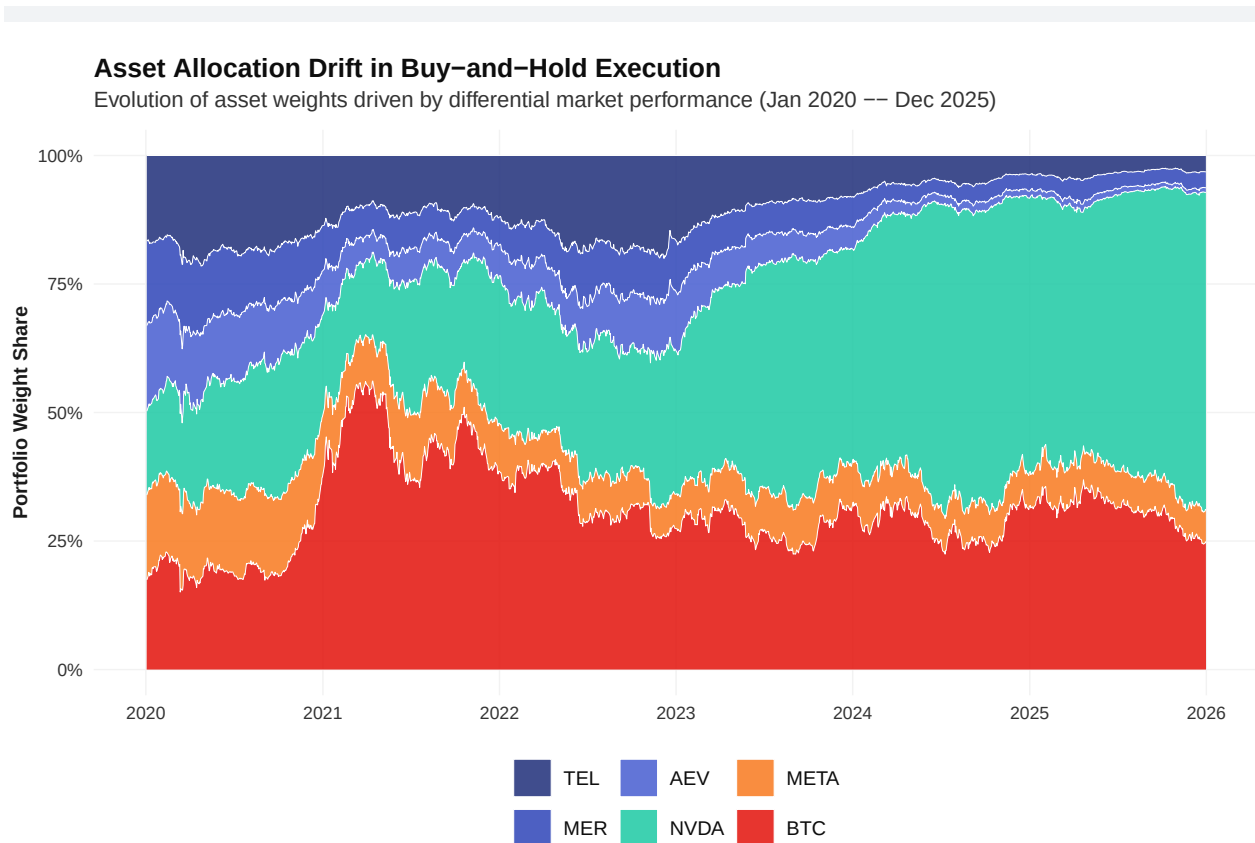


Figure 9: Asset Weight Drift Over Time in the Un-Rebalanced Buy-and-Hold Portfolio

7 Portfolio Optimization (Efficient Frontier, GMV, Max Sharpe)

```
mu_ann    <- colMeans(asset_returns_smp, na.rm = TRUE) * 252
Sigma_ann <- cov(asset_returns_smp, use = "pairwise.complete.obs") * 252

n_assets  <- length(mu_ann)
asset_names <- names(mu_ann)

# Equal-weight starting point for the solver
w0 <- rep(1 / n_assets, n_assets)
```

```
eval_f_var <- function(w) {
  as.numeric(t(w) %*% Sigma_ann %*% w)
}
```

```
eval_grad_f_var <- function(w) {
  nl.grad(w, eval_f_var)
```



```

}

eval_f_sharpe <- function(w) {
  port_ret <- sum(w * mu_ann)
  port_sd <- sqrt(as.numeric(t(w) %*% Sigma_ann %*% w))
  -(port_ret - rf_annual) / port_sd # negative since nloptr always
  ↪ minimizes
}

eval_grad_f_sharpe <- function(w) {
  nl.grad(w, eval_f_sharpe)
}

eval_g_eq_sum1 <- function(w) {
  sum(w) - 1
}

eval_jac_g_eq_sum1 <- function(w) {
  nl.jacobian(w, eval_g_eq_sum1)
}

# Solver settings: SLSQP as guided, since it's the R equivalent of Excel's
↪ GRG Nonlinear
opt_settings <- list(
  "algorithm" = "NLOPT_LD_SLSQP",
  "xtol_rel" = 1.0e-8,
  "maxeval" = 1000
)

gmw_solution <- nloptr(
  x0 = w0,
  eval_f = eval_f_var,
  eval_grad_f = eval_grad_f_var,
  lb = rep(0, n_assets), # long-only, no shorting
  ub = rep(1, n_assets), # no single asset above 100%
  eval_g_eq = eval_g_eq_sum1,
  eval_jac_g_eq = eval_jac_g_eq_sum1,
  opts = opt_settings
)

w_gmv <- gmw_solution$solution
names(w_gmv) <- asset_names

gmw_ret <- sum(w_gmv * mu_ann)
gmw_sd <- sqrt(as.numeric(t(w_gmv) %*% Sigma_ann %*% w_gmv))

```

```

sharpe_solution <- nloptr(
  x0          = w0,
  eval_f      = eval_f_sharpe,
  eval_grad_f = eval_grad_f_sharpe,
  lb          = rep(0, n_assets),
  ub          = rep(1, n_assets),
  eval_g_eq   = eval_g_eq_sum1,
  eval_jac_g_eq = eval_jac_g_eq_sum1,
  opts        = opt_settings
)

w_sharpe <- sharpe_solution$solution
names(w_sharpe) <- asset_names

sharpe_ret <- sum(w_sharpe * mu_ann)
sharpe_sd  <- sqrt(as.numeric(t(w_sharpe) %*% Sigma_ann %*% w_sharpe))
sharpe_val <- (sharpe_ret - rf_annual) / sharpe_sd

target_returns <- seq(min(mu_ann), max(mu_ann), length.out = 50)
frontier_list  <- vector("list", length(target_returns))

D <- 2 * Sigma_ann + diag(1e-6, n_assets)
d <- rep(0, n_assets)

for (i in seq_along(target_returns)) {
  target <- target_returns[i]

  Amat <- cbind(
    rep(1, n_assets), # sum(w) = 1
    mu_ann,           # sum(w * mu_ann) = target
    diag(n_assets),   # w >= 0
    -diag(n_assets)   # -w >= -1
  )
  bvec <- c(1, target, rep(0, n_assets), rep(-1, n_assets))

  sol <- tryCatch(
    quadprog::solve.QP(Dmat = D, dvec = d, Amat = Amat, bvec = bvec, meq =
→ 2),
    error = function(e) NULL
  )

  if (!is.null(sol)) {
    w_i <- sol$solution
    frontier_list[[i]] <- data.frame(

```

```

    Target_Return = target,
    Risk    = sqrt(as.numeric(t(w_i) %*% Sigma_ann %*% w_i)),
    Return = sum(w_i * mu_ann)
  )
}
}

efficient_frontier_df <- bind_rows(frontier_list)

efficient_only_df <- efficient_frontier_df[efficient_frontier_df$Return >=
  ↪  gm_v_ret, ]

ggplot(efficient_only_df, aes(x = Risk, y = Return)) +
  geom_path(color = "#1F2E7A", linewidth = 0.9) +
  geom_point(
    data = data.frame(Risk = gm_v_sd, Return = gm_v_ret),
    aes(x = Risk, y = Return), color = "#E3120B", size = 3
  ) +
  geom_point(
    data = data.frame(Risk = sharpe_sd, Return = sharpe_ret),
    aes(x = Risk, y = Return), color = "#1DC9A4", size = 3
  ) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Efficient Frontier: Risk-Return Tradeoff",
    subtitle = "Red point = Global Minimum Variance | Teal point = Maximum
  ↪  Sharpe Ratio",
    x = "Annualized Volatility (Risk)",
    y = "Annualized Expected Return"
  ) +
  theme_quant()

```

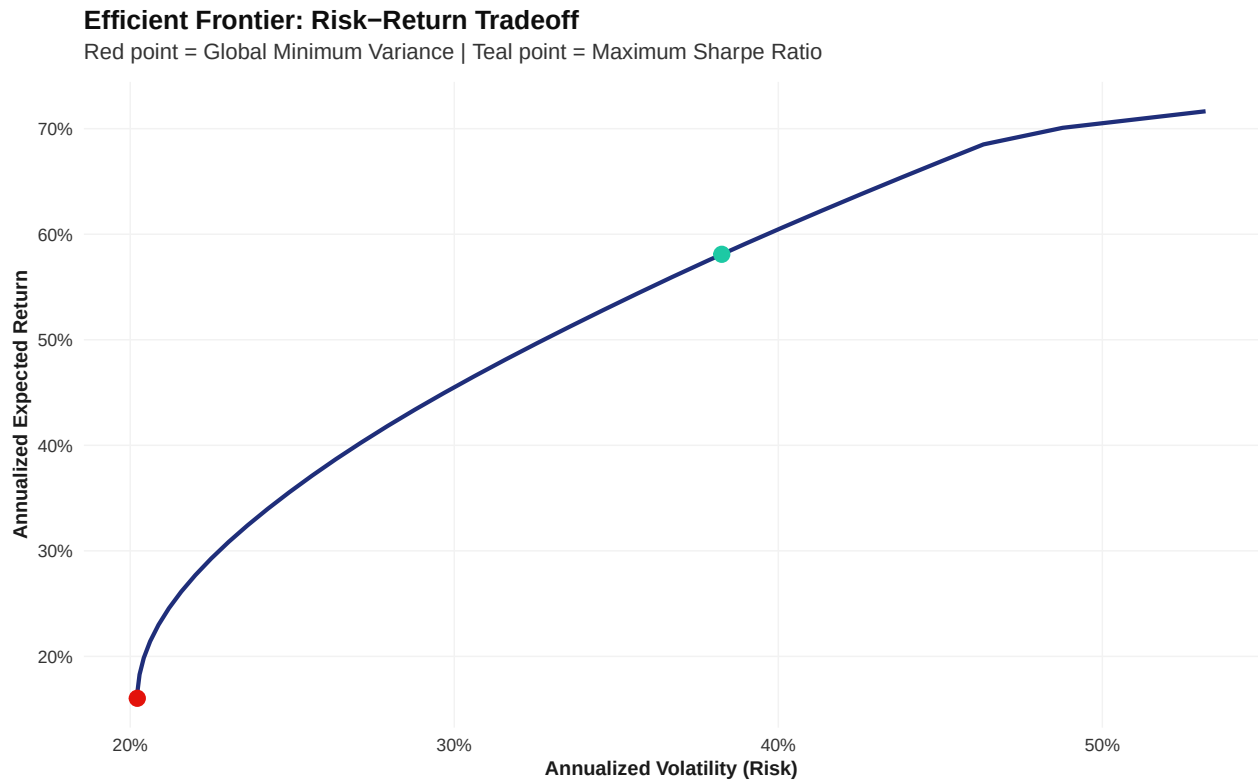


Figure 10: Efficient Frontier with Global Minimum Variance and Maximum Sharpe Portfolios

```
optimal_weights_df <- data.frame(
  Asset      = asset_names,
  GMV_Weight = sprintf("%.2f%%", w_gmv * 100),
  MaxSharpe_Weight = sprintf("%.2f%%", w_sharpe * 100)
)

safe_kable(optimal_weights_df, caption = "Optimal Portfolio Weights: GMV vs
↪ Maximum Sharpe")
```

Table 7: Optimal Portfolio Weights: GMV vs Maximum Sharpe

Asset	GMV_Weight	MaxSharpe_Weight
TEL	31.01%	0.00%
MER	31.23%	18.42%
AEV	16.85%	0.00%
NVDA	3.15%	55.44%
META	12.86%	0.00%
BTC	4.90%	26.14%

```
weights_long_df <- data.frame(
  Asset      = rep(asset_names, 2),
```

```

Portfolio = rep(c("GMV", "Max Sharpe"), each = n_assets),
Weight    = c(w_gmv, w_sharpe)
)

ggplot(weights_long_df, aes(x = Asset, y = Weight, fill = Asset)) +
  geom_col() +
  facet_wrap(~ Portfolio) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Optimal Portfolio Allocation by Asset",
    subtitle = "Comparison between Global Minimum Variance and Maximum
    ↪ Sharpe portfolios",
    x = NULL, y = "Portfolio Weight"
  ) +
  theme_quant() +
  theme(legend.position = "none")

```

Optimal Portfolio Allocation by Asset

Comparison between Global Minimum Variance and Maximum Sharpe portfolios

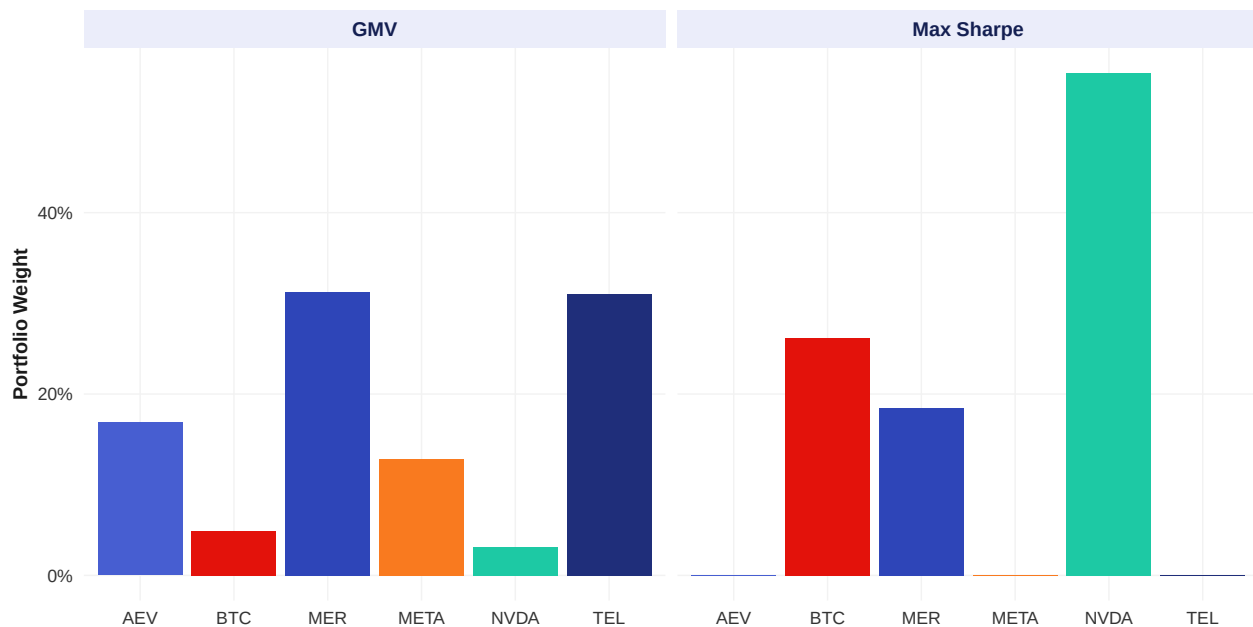


Figure 11: Optimal Portfolio Allocation by Asset (GMV vs Max Sharpe)

```

# Running-peak drawdown dates: depth measured from the running maximum
↪ wealth
# preceding the deepest trough, matching the reported max drawdown
↪ magnitudes.
dd_dates <- function(r_series) {

```

```

w <- cumprod(1 + as.numeric(r_series))
runmax <- cummax(w)
dd <- w / runmax - 1
t <- which.min(dd)
p <- which.max(w[1:t])
c(depth = sprintf("%.2f%%", dd[t] * 100), peak =
  → format(index(r_series)[p]), trough = format(index(r_series)[t]))
}

bnh_dd <- dd_dates(r_bnh)
reb_dd <- dd_dates(r_reb)

nvda_peak_w <- max(weights_bnh[, "NVDA"])
nvda_peak_d <- format(index(weights_bnh)[which.max(weights_bnh[, "NVDA"])])

ann_vol_sigma <- sqrt(diag(Sigma_ann))

cat("Buy & Hold max drawdown:", bnh_dd["depth"], "peak", bnh_dd["peak"],
  → "trough", bnh_dd["trough"], "\n")

```

Buy & Hold max drawdown: -58.21% peak 2021-11-08 trough 2022-10-14

```

cat("Monthly Rebalanced max drawdown:", reb_dd["depth"], "peak",
  → reb_dd["peak"], "trough", reb_dd["trough"], "\n")

```

Monthly Rebalanced max drawdown: -43.20% peak 2021-12-27 trough 2022-10-12

```

cat("Buy & Hold NVDA weight peak:", sprintf("%.2f%%", nvda_peak_w * 100),
  → "on", nvda_peak_d, "\n")

```

Buy & Hold NVDA weight peak: 62.29% on 2025-12-26

```

cat("Annualized volatility from optimization covariance (simple daily
  → returns x 252):", paste(sprintf("%s %.2f%%", names(ann_vol_sigma),
  → ann_vol_sigma * 100), collapse = ", "), "\n")

```

Annualized volatility from optimization covariance (simple daily returns x
 → 252): TEL 31.13%, MER 31.38%, AEV 38.91%, NVDA 53.18%, META 43.82%, BTC
 → 61.39%

8 Portfolio Risk Assessment (VaR, CVaR, Monte Carlo, Stress Testing)

```

port_returns_gmv <- Return.portfolio(

```

```

    R = asset_returns_smp,
    weights = w_gmv,
    rebalance_on = "months"
)

port_returns_sharpe <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_sharpe,
  rebalance_on = "months"
)

v_hist_gmv_95 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.95, method =
  ↪ "historical"))))
v_hist_gmv_99 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.99, method =
  ↪ "historical"))))
v_para_gmv_95 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.95, method =
  ↪ "gaussian"))))
v_para_gmv_99 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.99, method =
  ↪ "gaussian"))))

e_hist_gmv_95 <- abs(as.numeric(ES(port_returns_gmv, p = 0.95, method =
  ↪ "historical"))))
e_hist_gmv_99 <- abs(as.numeric(ES(port_returns_gmv, p = 0.99, method =
  ↪ "historical"))))
e_para_gmv_95 <- abs(as.numeric(ES(port_returns_gmv, p = 0.95, method =
  ↪ "gaussian"))))
e_para_gmv_99 <- abs(as.numeric(ES(port_returns_gmv, p = 0.99, method =
  ↪ "gaussian"))))

v_hist_shp_95 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.95, method =
  ↪ "historical"))))
v_hist_shp_99 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.99, method =
  ↪ "historical"))))
v_para_shp_95 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.95, method =
  ↪ "gaussian"))))
v_para_shp_99 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.99, method =
  ↪ "gaussian"))))

e_hist_shp_95 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.95, method =
  ↪ "historical"))))
e_hist_shp_99 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.99, method =
  ↪ "historical"))))
e_para_shp_95 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.95, method =
  ↪ "gaussian"))))

```

```

e_para_shp_99 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.99, method =
  ↪ "gaussian"))))

risk_summary_df <- data.frame(
  Portfolio = rep(c("GMV", "Max Sharpe"), each = 4),
  Metric    = rep(c("Historical VaR", "Parametric VaR", "Historical ES
  ↪ (CVaR)", "Parametric ES (CVaR)"), 2),
  `95% CI`  = sprintf("%.2f%%", c(
    v_hist_gmv_95, v_para_gmv_95, e_hist_gmv_95, e_para_gmv_95,
    v_hist_shp_95, v_para_shp_95, e_hist_shp_95, e_para_shp_95
  ) * 100),
  `99% CI`  = sprintf("%.2f%%", c(
    v_hist_gmv_99, v_para_gmv_99, e_hist_gmv_99, e_para_gmv_99,
    v_hist_shp_99, v_para_shp_99, e_hist_shp_99, e_para_shp_99
  ) * 100),
  check.names = FALSE
)

safe_kable(risk_summary_df, caption = "Historical and Parametric VaR / CVaR
  ↪ Summary (Daily, 95% and 99% Confidence)")

```

Table 8: Historical and Parametric VaR / CVaR Summary (Daily, 95% and 99% Confidence)

Portfolio	Metric	95% CI	99% CI
GMV	Historical VaR	1.80%	3.27%
GMV	Parametric VaR	2.03%	2.90%
GMV	Historical ES (CVaR)	2.85%	4.67%
GMV	Parametric ES (CVaR)	2.56%	3.33%
Max Sharpe	Historical VaR	3.39%	5.64%
Max Sharpe	Parametric VaR	3.74%	5.38%
Max Sharpe	Historical ES (CVaR)	5.11%	8.55%
Max Sharpe	Parametric ES (CVaR)	4.74%	6.19%

```

# Simulate 10,000 one-day return scenarios using multivariate normal
  ↪ distribution
set.seed(123)
n_sims <- 10000

mu_daily    <- mu_ann / 252
Sigma_daily <- Sigma_ann / 252

sim_asset_returns <- mvrnorm(n = n_sims, mu = mu_daily, Sigma = Sigma_daily)
colnames(sim_asset_returns) <- asset_names

sim_port_returns_gmv <- as.numeric(sim_asset_returns %*% w_gmv)

```



```

sim_port_returns_sharpe <- as.numeric(sim_asset_returns %*% w_sharpe)

mc_var_gmv_95      <- quantile(sim_port_returns_gmv, 0.05)
mc_var_sharpe_95 <- quantile(sim_port_returns_sharpe, 0.05)

mc_es_gmv_95      <- mean(sim_port_returns_gmv[sim_port_returns_gmv <=
  ↪ mc_var_gmv_95])
mc_es_sharpe_95 <- mean(sim_port_returns_sharpe[sim_port_returns_sharpe <=
  ↪ mc_var_sharpe_95])

mc_summary_df <- data.frame(
  Portfolio = c("GMV", "Max Sharpe"),
  `MC VaR (95%)` = sprintf("%.2f%%", c(abs(mc_var_gmv_95),
  ↪ abs(mc_var_sharpe_95)) * 100),
  `MC ES (95%)` = sprintf("%.2f%%", c(abs(mc_es_gmv_95),
  ↪ abs(mc_es_sharpe_95)) * 100),
  check.names = FALSE
)

safe_kable(mc_summary_df, caption = "Monte Carlo Simulated VaR and Expected
  ↪ Shortfall (10,000 Simulations)")

```

Table 9: Monte Carlo Simulated VaR and Expected Shortfall (10,000 Simulations)

Portfolio	MC VaR (95%)	MC ES (95%)
GMV	2.02%	2.54%
Max Sharpe	3.68%	4.72%

```

dist_compare_df <- bind_rows(
  data.frame(Return = as.numeric(port_returns_sharpe), Source =
    ↪ "Historical"),
  data.frame(Return = sim_port_returns_sharpe, Source = "Monte Carlo")
)

ggplot(dist_compare_df, aes(x = Return, fill = Source, color = Source)) +
  geom_density(alpha = 0.35, linewidth = 0.7) +
  scale_fill_manual(values = c("Historical" = "#1F2E7A", "Monte Carlo" =
    ↪ "#E3120B")) +
  scale_color_manual(values = c("Historical" = "#1F2E7A", "Monte Carlo" =
    ↪ "#E3120B")) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Historical vs Monte Carlo Simulated Return Distribution",
    subtitle = "Max Sharpe Portfolio: comparing observed daily returns to
    ↪ 10,000 simulated draws",

```

```

x = "Daily Portfolio Return",
y = "Density"
) +
theme_quant()

```

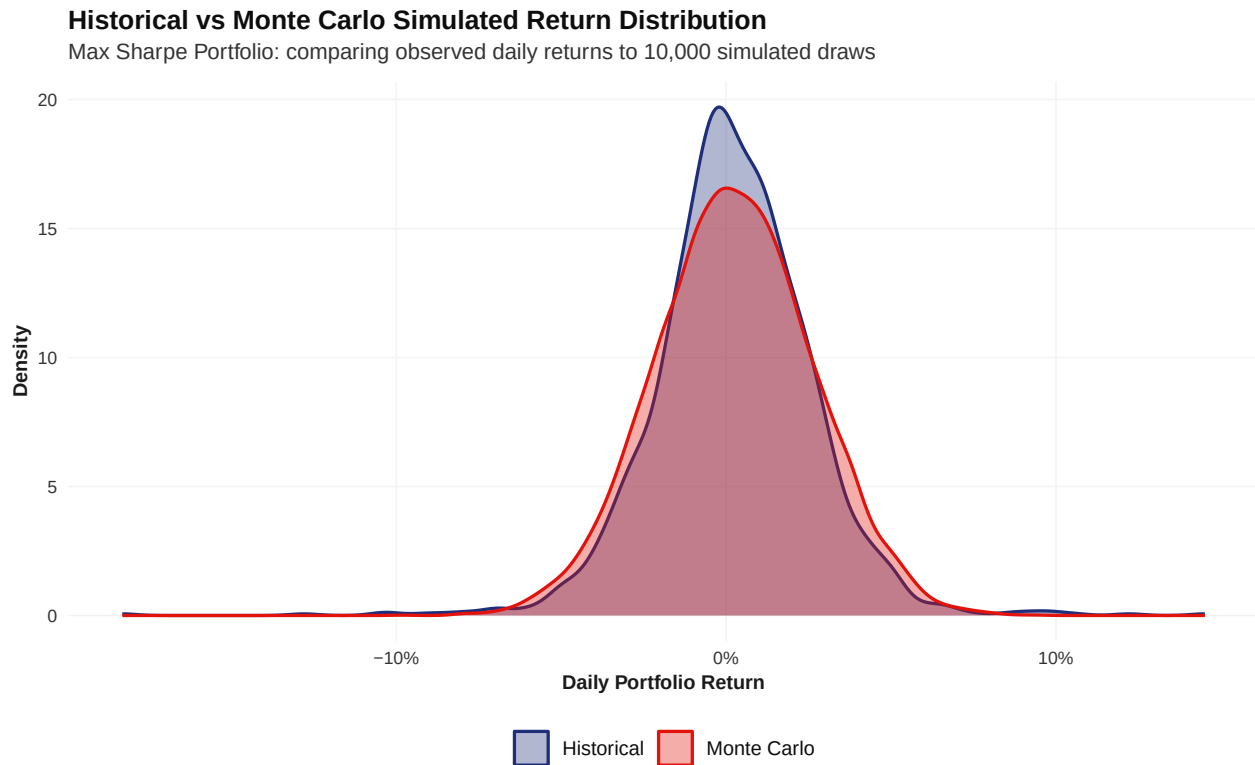


Figure 12: Historical vs Monte Carlo Simulated Return Distribution (Max Sharpe Portfolio)

```

# Historical Stress Scenario Windows
covid_crash_window    <- "2020-02-19/2020-03-23"
tech_selloff_window   <- "2021-11-19/2022-01-27"
crypto_winter_window  <- "2022-05-05/2022-05-13"
fed_hike_window       <- "2022-06-08/2022-06-16"

calc_window_shock <- function(ret_xts, date_range) {
  window_returns <- ret_xts[date_range]
  sapply(window_returns, function(col) prod(1 + col, na.rm = TRUE) - 1)
}

shock_covid    <- calc_window_shock(ret_smp, covid_crash_window)
shock_tech     <- calc_window_shock(ret_smp, tech_selloff_window)
shock_crypto   <- calc_window_shock(ret_smp, crypto_winter_window)
shock_fed      <- calc_window_shock(ret_smp, fed_hike_window)

```

```

worst_observed <- pmin(shock_covid, shock_crypto, shock_fed, shock_tech)
shock_black_swan <- worst_observed * 1.5

stress_scenarios <- tibble(
  Scenario = c("2020 COVID Crash", "2022 Crypto Winter", "Fed Rate Hike
    → Shock",
              "Tech Sector Selloff", "Hypothetical Black Swan"),
  TEL = c(shock_covid["TEL"], shock_crypto["TEL"], shock_fed["TEL"],
    → shock_tech["TEL"], shock_black_swan["TEL"]),
  MER = c(shock_covid["MER"], shock_crypto["MER"], shock_fed["MER"],
    → shock_tech["MER"], shock_black_swan["MER"]),
  AEV = c(shock_covid["AEV"], shock_crypto["AEV"], shock_fed["AEV"],
    → shock_tech["AEV"], shock_black_swan["AEV"]),
  NVDA = c(shock_covid["NVDA"], shock_crypto["NVDA"], shock_fed["NVDA"],
    → shock_tech["NVDA"], shock_black_swan["NVDA"]),
  META = c(shock_covid["META"], shock_crypto["META"], shock_fed["META"],
    → shock_tech["META"], shock_black_swan["META"]),
  BTC = c(shock_covid["BTC"], shock_crypto["BTC"], shock_fed["BTC"],
    → shock_tech["BTC"], shock_black_swan["BTC"])
)

```

```

stress_impact_df <- stress_scenarios %>%
  rowwise() %>%
  mutate(
    shock_vec = list(c(TEL, MER, AEV, NVDA, META, BTC)),
    GMV_Impact = sum(unlist(shock_vec) *
    → w_gmv[c("TEL", "MER", "AEV", "NVDA", "META", "BTC")]),
    MaxSharpe_Impact = sum(unlist(shock_vec) *
    → w_sharpe[c("TEL", "MER", "AEV", "NVDA", "META", "BTC")])
  ) %>%
  ungroup() %>%
  dplyr::select(Scenario, GMV_Impact, MaxSharpe_Impact)

```

```

stress_summary_df <- stress_impact_df %>%
  mutate(
    GMV_Impact = sprintf("%.2f%%", GMV_Impact * 100),
    MaxSharpe_Impact = sprintf("%.2f%%", MaxSharpe_Impact * 100)
  ) %>%
  rename(`GMV Portfolio Impact` = GMV_Impact, `Max Sharpe Portfolio Impact`
    → = MaxSharpe_Impact)

safe_kable(stress_summary_df, caption = "Portfolio Impact Under Stress
  → Scenarios")

```

Table 10: Portfolio Impact Under Stress Scenarios

Scenario	GMV Portfolio Impact	Max Sharpe Portfolio Impact
2020 COVID Crash	-22.55%	-28.83%
2022 Crypto Winter	-4.90%	-14.40%
Fed Rate Hike Shock	-9.47%	-18.65%
Tech Sector Selloff	3.71%	-24.58%
Hypothetical Black Swan	-35.89%	-45.30%

```

stress_long_df <- stress_impact_df %>%
  pivot_longer(cols = c(GMV_Impact, MaxSharpe_Impact), names_to =
    ↪ "Portfolio", values_to = "Impact") %>%
  mutate(Portfolio = ifelse(Portfolio == "GMV_Impact", "GMV", "Max Sharpe"))

ggplot(stress_long_df, aes(x = reorder(Scenario, Impact), y = Impact, fill
  ↪ = Portfolio)) +
  geom_col(position = "dodge") +
  coord_flip() +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_fill_manual(values = c("GMV" = "#1F2E7A", "Max Sharpe" = "#E3120B"))
  ↪ +
  labs(
    title = "Portfolio Impact Under Stress Scenarios",
    subtitle = "Simulated instantaneous shock to portfolio value, GMV vs Max
      ↪ Sharpe weighting",
    x = NULL,
    y = "Portfolio Return Impact"
  ) +
  theme_quant()

```

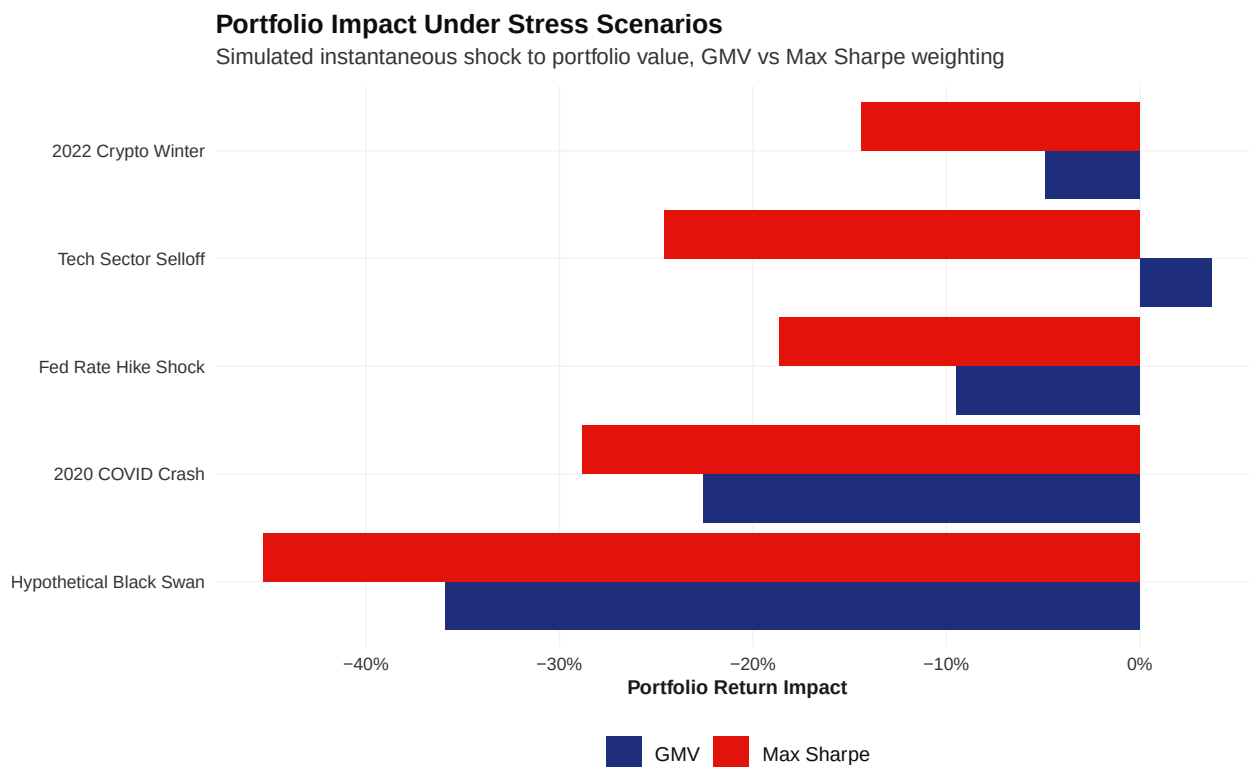


Figure 13: Portfolio Value Impact Across Stress Scenarios

9 Executive Investment Recommendation

Based on the portfolio optimization, historical performance, downside-risk measures, and stress-testing results, the Maximum Sharpe Ratio portfolio is recommended. The portfolio allocates 55.44 percent to NVIDIA (NVDA), 26.14 percent to Bitcoin (BTC), and 18.42 percent to Manila Electric Company (MER), delivering an expected annual return of 58.10 percent, an annualized volatility of 38.25 percent, and a Sharpe ratio of 1.3815 at the 5.25 percent risk-free rate, the highest risk-adjusted profile in the feasible universe. Figure 11 contrasts this allocation with the GMV portfolio, and Figure 10 locates it at the tangency point of the efficient frontier.

NVIDIA contributes the strongest sample-period expected return (71.65 percent annualized) and, together with Bitcoin (61.49 percent annualized), drives the portfolio's return profile. Bitcoin's inclusion rests on its high expected return and its low correlation with the Philippine constituents (MER–BTC 0.0286), while NVIDIA pairs with the domestic block at near-zero correlation (MER–NVDA 0.0121, MER–META 0.0172) and at moderate co-movement with Bitcoin (NVDA–BTC 0.3204). Manila Electric Company anchors the allocation as the sole domestic constituent, combining a positive expected return (12.49 percent annualized) with near-zero correlation to both growth assets. TEL, AEV, and META receive zero weight in the long-only tangency solution.

Downside risk is governed by the empirical tail measures reported in Table 8. The recommended portfolio’s daily 95 percent Historical VaR is -3.39 percent and its 95 percent Historical Expected Shortfall is -5.11 percent; at the 99 percent level these reach -5.64 percent and -8.55 percent. The empirical 99 percent Expected Shortfall (8.55 percent) exceeds the Gaussian parametric estimate (6.19 percent) by 236 basis points per day, so historical measures, rather than the normal approximation, should set the risk budget.

Stress testing confirms the trade-off between return and capital preservation. Under the five scenarios in Figure 13 and Table 10, the Maximum Sharpe portfolio loses 28.83 percent in the 2020 COVID crash, 24.58 percent in the technology-sector selloff, 18.65 percent in the 2022 rate-hike shock, and 14.40 percent in the 2022 crypto winter, with a -45.30 percent loss under the hypothetical 1.5x Black Swan scenario. The GMV portfolio contains the same shocks at materially smaller magnitudes (-22.55 percent COVID, -35.89 percent Black Swan), illustrating its role as the defensive alternative.

Disciplined monthly rebalancing is required to manage the portfolio’s concentration: NVDA and BTC together represent 81.58 percent of the recommended allocation, and un-rebalanced equal-weight exposure drifted to 61.75 percent NVDA and 24.77 percent BTC by December 2025. Over the sample, monthly rebalancing reduced volatility from 35.12 percent to 24.18 percent and the peak drawdown from -58.21 percent to -43.20 percent at the cost of cumulative return (384.85 percent versus 743.04 percent). For investors prioritizing capital preservation, the GMV portfolio allocates TEL 31.01 percent, MER 31.23 percent, AEV 16.85 percent, NVDA 3.15 percent, META 12.86 percent, and BTC 4.90 percent, delivering a 16.03 percent expected return at 20.22 percent volatility (Sharpe 0.5332). Portfolio choice ultimately reflects the investor’s willingness and capacity to bear losses: the Maximum Sharpe portfolio suits investors seeking higher returns despite greater risk, while the GMV portfolio suits investors who cannot tolerate peak stress losses near -45 percent.

10 References

- Artzner, P., Delbaen, F., Eber, J. M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203-228.
- Bodie, Z., Kane, A., & Marcus, A. J. (2021). *Investments* (12th ed.). McGraw-Hill Education.
- Fabozzi, F. J., Focardi, S. M., & Kolm, P. N. (2002). *Financial Modeling of the Equity Market: From Portfolio Optimization to Stress Testing*. John Wiley & Sons.
- Jorion, P. (2007). *Value at Risk: The New Benchmark for Managing Financial Risk* (3rd ed.). McGraw-Hill.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91.
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425-442.

11 Appendix: Complete R Analysis Code

```
# FINSTAD Case 2 Group 3 (Charlie) Complete Portfolio & Risk Pipeline

library(tidyverse)
library(lubridate)
library(scales)
library(xts)
library(zoo)
library(PerformanceAnalytics)
library(nloptr)
library(quadprog)
library(MASS)

raw_data <- read_csv("../data/master_dataset.csv", show_col_types = FALSE)
→ %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2020-01-02") & Date <= as.Date("2025-12-31")) %>%
  arrange(Date)

asset_cols <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC", "SPY")
portfolio_assets <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC")

prices_mat <- as.matrix(raw_data[, asset_cols])
prices_xts <- xts(prices_mat, order.by = raw_data[["Date"]])
prices_carried <- na.locf(prices_xts)

returns_xts_carried <- Return.calculate(prices_carried, method = "log")[-1,
→ ]
asset_returns_carried <- returns_xts_carried[, portfolio_assets]

ret_smp <- exp(returns_xts_carried) - 1
asset_returns_smp <- ret_smp[, portfolio_assets]

w_eq <- rep(1 / 6, 6)
names(w_eq) <- portfolio_assets

port_bnh <- Return.portfolio(R = asset_returns_smp, weights = w_eq)
port_reb <- Return.portfolio(R = asset_returns_smp, weights = w_eq,
→ rebalance_on = "months")

mu_ann <- colMeans(asset_returns_smp, na.rm = TRUE) * 252
Sigma_ann <- cov(asset_returns_smp, use = "pairwise.complete.obs") * 252
rf_annual <- 0.0525
```

```

eval_f_var <- function(w) as.numeric(t(w) %*% Sigma_ann %*% w)
eval_grad_f_var <- function(w) nl.grad(w, eval_f_var)
eval_g_eq <- function(w) sum(w) - 1
eval_jac_g_eq <- function(w) nl.jacobian(w, eval_g_eq)

w0 <- rep(1/6, 6)
gmw_sol <- nloptr(x0 = w0, eval_f = eval_f_var, eval_grad_f =
  ↪ eval_grad_f_var,
               lb = rep(0,6), ub = rep(1,6), eval_g_eq = eval_g_eq,
               eval_jac_g_eq = eval_jac_g_eq,
               opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel" =
  ↪ 1e-8))
w_gmw <- gmw_sol$solution
names(w_gmw) <- portfolio_assets

eval_f_sharpe <- function(w) {
  p_ret <- sum(w * mu_ann)
  p_sd <- sqrt(as.numeric(t(w) %*% Sigma_ann %*% w))
  -(p_ret - rf_annual) / p_sd
}
eval_grad_f_sharpe <- function(w) nl.grad(w, eval_f_sharpe)

shp_sol <- nloptr(x0 = w0, eval_f = eval_f_sharpe, eval_grad_f =
  ↪ eval_grad_f_sharpe,
               lb = rep(0,6), ub = rep(1,6), eval_g_eq = eval_g_eq,
               eval_jac_g_eq = eval_jac_g_eq,
               opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel" =
  ↪ 1e-8))
w_sharpe <- shp_sol$solution
names(w_sharpe) <- portfolio_assets

port_ret_gmw <- Return.portfolio(R = asset_returns_smp, weights = w_gmw,
  ↪ rebalance_on = "months")
port_ret_shp <- Return.portfolio(R = asset_returns_smp, weights = w_sharpe,
  ↪ rebalance_on = "months")

var_gmw_95 <- abs(VaR(port_ret_gmw, p = 0.95, method = "historical"))
var_shp_95 <- abs(VaR(port_ret_shp, p = 0.95, method = "historical"))
es_gmw_95 <- abs(ES(port_ret_gmw, p = 0.95, method = "historical"))
es_shp_95 <- abs(ES(port_ret_shp, p = 0.95, method = "historical"))

```